

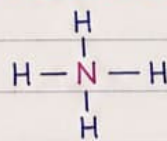
## Chapter - 9

# AMINES

### NEET Point

- Basic Nature
- Structure
- Reactions
- Importance

**Definition:** Amines are organic compounds derived by the replacement of one or more hydrogen atoms of ammonia ( $\text{NH}_3$ ) by alkyl or aryl groups.

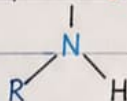
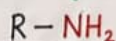


Ammonia

## 1. CLASSIFICATION OF AMINES

Amines are classified based on the number of alkyl or aryl groups attached to the nitrogen atom.

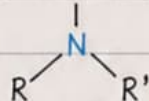
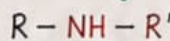
### Primary ( $1^\circ$ )



One alkyl/aryl group

Example:  $\text{CH}_3\text{NH}_2$   
Methylamine

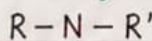
### Secondary ( $2^\circ$ )



Two alkyl/aryl groups

Example:  $(\text{CH}_3)_2\text{NH}$   
Dimethylamine

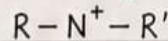
### Tertiary ( $3^\circ$ )



Three alkyl/aryl groups

Example:  $(\text{CH}_3)_3\text{N}$   
Trimethylamine

### Quaternary ( $4^\circ$ )



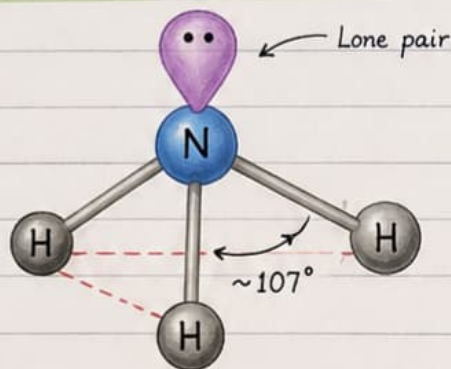
$\text{X}^-$

Four alkyl/aryl groups

Example:  $(\text{CH}_3)_4\text{N}^+\text{Cl}^-$   
Tetramethylammonium chloride

## 2. STRUCTURE OF AMINES

- Nitrogen in amines is  $\text{sp}^3$  hybridised.
- The shape is **trigonal pyramidal**.
- Bond angle is slightly less than  $109.5^\circ$  due to lone pair-bond pair repulsion.
- Lone pair on nitrogen is responsible for **basic nature**.



Trigonal pyramidal structure

## 3. NOMENCLATURE (BASICS)

- Primary amines: Alkyl/aryl name + amine
- Secondary amines: N-alkyl name + alkyl + amine
- Tertiary amines: N,N-dialkyl name + alkyl + amine

### Examples:

$\text{CH}_3\text{NH}_2 \rightarrow$  Methanamine (Methylamine)

$\text{C}_2\text{H}_5\text{NH}_2 \rightarrow$  Ethanamine (Ethylamine)

$(\text{CH}_3)_2\text{NH} \rightarrow$  Dimethylamine

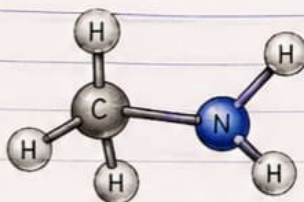
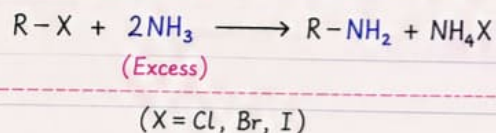
$(\text{C}_2\text{H}_5)(\text{CH}_3)_2\text{N} \rightarrow$  N,N-Dimethyl ethanamine

(More on Next Page  $\rightarrow$ )

## 4. PREPARATION OF AMINES

### ① From Ammonia (NH<sub>3</sub>)

- Alkyl or aryl halides react with excess NH<sub>3</sub>.



Ethylamine (C<sub>2</sub>H<sub>5</sub>NH<sub>2</sub>)

#### Key Point

- Alkyl halides give 1° amines with excess NH<sub>3</sub>.
- Aryl halides are less reactive.

### ② By Reduction of Nitro Compounds

- Nitro compounds on reduction give primary amines.

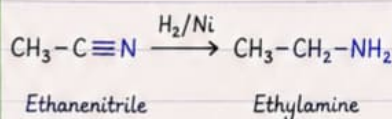


Nitrobenzene

Aniline

### ③ By Reduction of Nitriles

- Nitriles on reduction give primary amines.

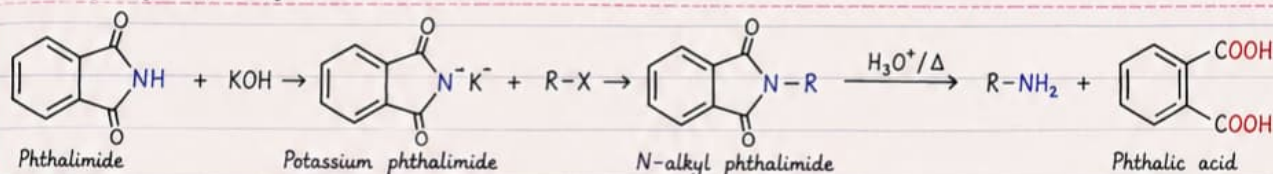


#### Exam Tip

Reduction of -NO<sub>2</sub> or -CN group is a common method to prepare amines.

### ④ By Gabriel Phthalimide Synthesis

- Alkyl halide reacts with potassium phthalimide.
- On hydrolysis, primary amine is obtained.



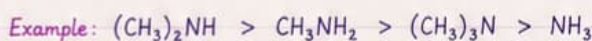
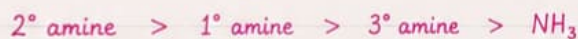
## 5. CHEMICAL PROPERTIES OF AMINES

### (A) Basic Nature

- Amines act as Lewis bases due to lone pair on nitrogen.
- They accept proton from acids to form ammonium salts.

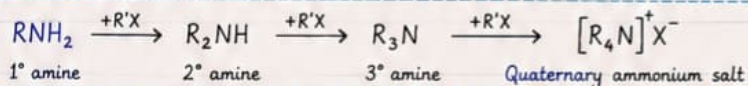


#### Basic Strength Order (in aqueous solution)

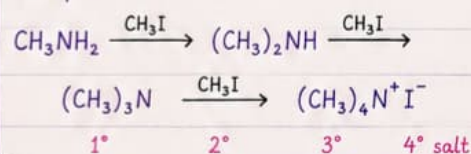


### (B) Alkylation

- Amines react with alkyl halides to form higher (2°, 3°) amines and ultimately quaternary ammonium salts.



Example:



### (C) Acylation (Hinsberg Test)

- Amines react with benzenesulphonyl chloride (C<sub>6</sub>H<sub>5</sub>SO<sub>2</sub>Cl) to give sulphonamides (Hinsberg test is used to distinguish 1°, 2° and 3° amines).

|          |                                                                                    |                                        |
|----------|------------------------------------------------------------------------------------|----------------------------------------|
| 1° amine | $R-NH_2 + C_6H_5SO_2Cl \longrightarrow R-NH-SO_2C_6H_5 + HCl$<br>Soluble in alkali | $R-\overset{O}{\parallel}N-SO_2C_6H_5$ |
| 2° amine | $R_2NH + C_6H_5SO_2Cl \longrightarrow R_2N-SO_2C_6H_5 + HCl$<br>Soluble in alkali  | $R-\overset{O}{\parallel}N-SO_2C_6H_5$ |
| 3° amine | $R_3N + C_6H_5SO_2Cl \longrightarrow \text{No reaction}$<br>Insoluble in alkali    | —                                      |

#### Remember!

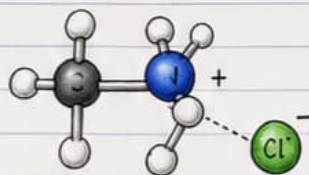
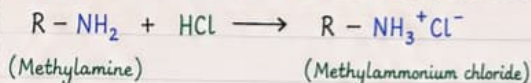
Hinsberg test is a classic method to identify the type of amine.

Amines are important in medicines, dyes, polymers & many natural products.

## 6. REACTIONS OF AMINES

### (A) With Acids

Amines act as **Lewis bases** and react with acids to form ammonium salts.



Methylammonium chloride

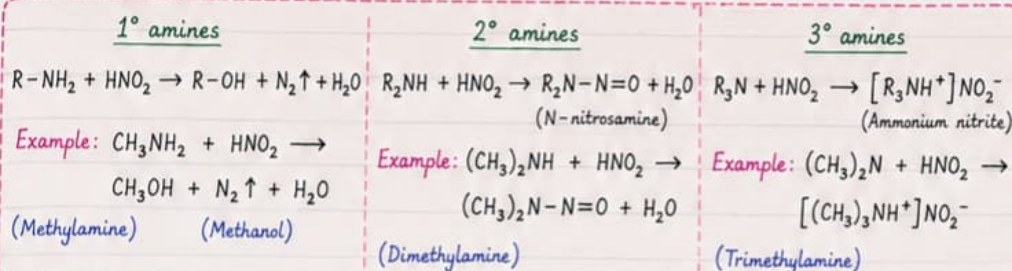
### Quick Note

Amines are less reactive than ammonia because alkyl/aryl groups donate electrons to nitrogen.



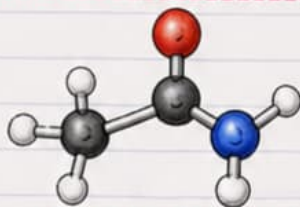
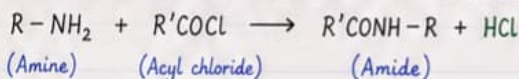
### (B) With Nitrous Acid ( $\text{HNO}_2$ )

- 1° aliphatic amines  $\rightarrow$  give alcohol + nitrogen gas.
- 2° aliphatic amines  $\rightarrow$  give N-nitrosamines.
- 3° aliphatic amines  $\rightarrow$  form ammonium nitrite (no  $\text{N}_2$  gas).



### (C) With Acyl Chlorides

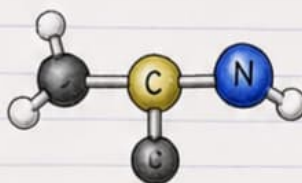
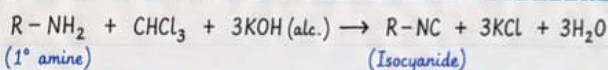
- Amines react with acyl chlorides to form amides.



Acetamide ( $\text{CH}_3\text{CONH}_2$ )

### (D) Carbylamine Reaction (Only 1° amines)

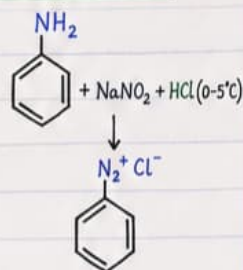
- 1° amines react with chloroform and alcoholic KOH to form foul-smelling isocyanides (carbylamines).
- 2° and 3° amines do not give this reaction.



Isocyanide ( $\text{R-NC}$ )

### Important!

Aromatic 1° amines do not give  $\text{N}_2$  gas. Instead they form **diazonium salts**.



Benzenediazonium chloride

### Remember

1° amines are the most reactive towards nitrous acid and acylation.

## 7. BASIC STRENGTH OF AMINES

| Type of amine    | Basic strength (in aqueous solution)                                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Aliphatic amines | $2^\circ > 1^\circ > 3^\circ > \text{NH}_3$                                                                                          |
| Aromatic amines  | Aliphatic amines > Aniline ( $\text{C}_6\text{H}_5\text{NH}_2$ )                                                                     |
| Reason           | Inductive effect (+I) of alkyl groups increases basicity. In 3°, steric hindrance & poor solvation of ammonium ion reduces basicity. |

### Factors affecting basicity

- +I effect of alkyl groups  $\uparrow$  basicity
- Steric hindrance  $\downarrow$  basicity
- Solvation of ammonium ion  $\uparrow$  basicity
- Resonance (as in aniline)  $\downarrow$  basicity

## 8. OCCURRENCE AND IMPORTANCE OF AMINES

### Occurrence

- A simple amine, methylamine, is found in decomposing fish.
- Many 2° amines occur in plants.
- Aniline occurs in coal tar.
- Amines are present in many natural products like alkaloids, amino acids, proteins, etc.

### Importance / Uses

- Used in the manufacture of dyes, drugs, plastics, pesticides, rubber chemicals.
- Aniline dyes are extensively used in textiles and leather industry.
- Many drugs (e.g., antihistamines, local anaesthetics) contain amine group.

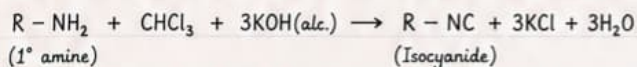
### Example - Natural Alkaloid



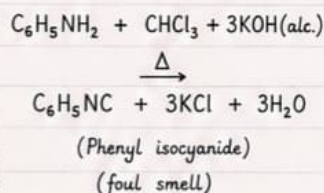
## 9. IDENTIFICATION OF AMINES

### (A) Carbylamine Test (Only 1° amines)

- 1° amines on heating with chloroform and alcoholic KOH give foul-smelling isocyanides (carbylamines).
- 2° and 3° amines do not respond.



#### Example (Aniline)



#### Key Point

Carbylamine test is a specific test for 1° amines.

### (B) Hinsberg Test (Distinguishes 1°, 2° and 3° amines)

- Amines are treated with benzenesulphonyl chloride ( $C_6H_5SO_2Cl$ ) in the presence of NaOH.

| Type of amine | Reaction with $C_6H_5SO_2Cl$                                                                                                                                          | In NaOH                                                                                                | In dil. HCl                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1° amine      | $C_6H_5-\overset{\overset{O}{\parallel}}{S}-Cl + R-NH_2 \rightarrow C_6H_5-\overset{\overset{O}{\parallel}}{S}-NH-R + HCl$ <p>(N-alkyl benzene sulphonamide)</p>      | <p>Soluble<br/>(forms salt)</p> $C_6H_5-\overset{\overset{O}{\parallel}}{S}-N^--R \cdot ONa$           | <p>Insoluble<br/>(precipitates)</p> $C_6H_5-\overset{\overset{O}{\parallel}}{S}-NH-R$ |
| 2° amine      | $C_6H_5-\overset{\overset{O}{\parallel}}{S}-Cl + R_2NH \rightarrow C_6H_5-\overset{\overset{O}{\parallel}}{S}-N(R)-H + HCl$ <p>(N,N-dialkyl benzene sulphonamide)</p> | <p>Insoluble<br/>(does not form salt)</p> $C_6H_5-\overset{\overset{O}{\parallel}}{S}-N(R)-H$          | <p>Insoluble<br/>(does not dissolve)</p>                                              |
| 3° amine      | $C_6H_5-\overset{\overset{O}{\parallel}}{S}-Cl + R_3N \rightarrow [C_6H_5-SO_2]^- + R_3N-H^+Cl^-$ <p>(no sulphonamide)</p>                                            | <p>Soluble<br/>(forms salt)</p> <p><math>R_3N</math> (no reaction with benzene sulphonyl chloride)</p> | <p>Soluble<br/>(forms salt)</p> <p><math>R_3N-H^+Cl^-</math></p>                      |

#### Note

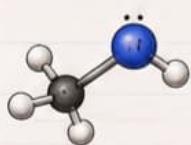
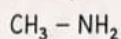
Hinsberg test is used to identify the class of amines.

#### Common Example

- Methylamine  $\rightarrow$  1°
- Dimethylamine  $\rightarrow$  2°
- Trimethylamine  $\rightarrow$  3°

## 10. SOME IMPORTANT AMINES

### (1) Methylamine

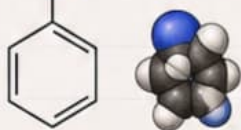


- Used in the manufacture of pharmaceuticals, dyes, rubber chemicals.

Stronger base than  $NH_3$



### (2) Aniline

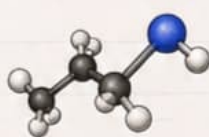
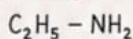


- Used in dyes, polymers (PU), explosives.
- Starting material for many industries.

Weaker base than  $NH_3$



### (3) Ethylamine

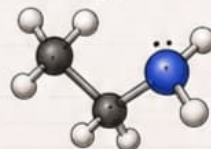
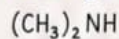


- Used in pharmaceuticals, agrochemicals.
- Intermediate in organic synthesis.

Stronger base than  $NH_3$



### (4) Dimethylamine



- Used in dyes, rubber chemicals, pharmaceuticals.
- Used as a solvent and in organic synthesis.

Stronger base than  $NH_3$



#### Compared Basicity (in aqueous solution)

2° amine > 1° amine > 3° amine >  $NH_3$

#### Reason:

2° amine has +I effect and good solvation.



## 11. SUMMARY

- Amines are derivatives of ammonia where one or more H atoms are replaced by alkyl/aryl groups.
- Classified as 1°, 2°, 3° and 4° (quaternary ammonium salts).
- Prepared by various methods such as reduction of nitro, nitriles, Gabriel phthalimide synthesis, etc.
- React as bases, form salts with acids and undergo characteristic reactions.
- Identification is done by Carbylamine test and Hinsberg test.
- Amines are widely used in industries and medicines.

Keep Practicing and Stay Curious!



"Chemistry is not just a subject, it's a way of understanding nature."

